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Signal representation and load transfer in bearing fault classification

Representación de señal y transferencia entre cargas en fallas de rodamientos

Abstract

Comparisons of signal representations for bearing fault diagnosis are usually reported under a single operating condition, where accuracy saturates and the comparison carries little information. This work measures how four representations behave when the test load differs from the training load. Time-domain statistics, FFT magnitude, db4 wavelet band energies and STFT spectrogram were combined with three vector classifiers over the sixteen (training load, test load) pairs of the CWRU drive-end dataset at 12 kHz, with five folds per cell and a fixed seed. The split was performed by source record and by purged contiguous blocks, so that no two windows sharing samples fall on opposite sides. Within a single load, where the split shares records, all four exceeded 0.95 macro-F1, so the usual comparison does not discriminate between them. Across loads the mean drop ranged from 0.074 for the spectrogram to 0.153 for time-domain statistics, with overlapping bootstrap intervals. The omnibus Friedman test over the twelve off-diagonal pairs did not reject, while the three per-classifier tests each did and disagreed on the ranking. An exploratory three-way ANOVA on the cell means places the representation by classifier interaction above every main effect. An amplitude-normalisation ablation did not improve transfer and degraded it with Random Forest. Representation choice interacts with the classifier rather than defining a general ordering.

Keywords: bearing fault diagnosis, data leakage, domain shift, Friedman test, signal representation, vibration analysis

Resumen

Las comparaciones entre representaciones de señal para el diagnóstico de fallas en rodamientos suelen reportarse bajo una sola condición de operación, donde la exactitud satura y aporta poca información. Este trabajo mide el comportamiento de cuatro representaciones cuando la carga de prueba difiere de la de entrenamiento. Se combinaron estadísticos temporales, magnitud FFT, energías de banda wavelet db4 y espectrograma STFT con tres clasificadores de vector sobre los dieciséis pares (carga de entrenamiento, carga de prueba) del conjunto CWRU a 12 kHz, con cinco pliegues por celda y semilla fija. La división se hizo por registro de origen y por bloques contiguos purgados, evitando fuga por solape. Dentro de una misma carga, donde la partición comparte registros, las cuatro superaron 0,95 de F1-macro, de modo que la comparación habitual no discrimina. Entre cargas la caída media osciló entre 0,074 para el espectrograma y 0,153 para los estadísticos, con intervalos de confianza solapados. La prueba de Friedman ómnibus sobre los doce pares fuera de la diagonal no rechazó, mientras que las tres pruebas por clasificador rechazaron y discreparon en el orden. Un ANOVA exploratorio sitúa la interacción representación-clasificador por encima de todo efecto principal. Una ablación de normalización de amplitud no mejoró la transferencia y la degradó con Random Forest. La elección de representación interactúa con el clasificador en lugar de definir un orden general.

Palabras clave: análisis de vibraciones, desplazamiento de dominio, diagnóstico de fallas en rodamientos, fuga de datos, prueba de Friedman, representación de señal

1. Introduction

Rolling element bearings fail in ways that leave a signature in the vibration measured on the housing. A localised defect produces an impact each time a rolling element passes over it, and those impacts excite the structural resonance of the assembly at a repetition rate set by the geometry and the shaft speed, which is the basis of envelope analysis and of the diagnostic techniques reviewed in [1].

Data-driven diagnosis adds a decision on top of that physics: the signal is converted into some representation and a classifier is trained on it. The choice of representation is normally justified by comparing several candidates on the same data and recommending the most accurate. The dataset of the Case Western Reserve University Bearing Data Center [2] is the standard reference for those experiments, to the point that a benchmark study was devoted to categorising it and recommending how it should be used [3], with later work proposing further improvements to that practice [4].

Accuracy within a single operating condition saturates near unity, so the ranking it produces separates methods by hundredths of a point and does not survive a change of dataset, and the way the data are partitioned decides the result: bearing vibration is usually cut into overlapping windows, and if those windows are shuffled before splitting, windows sharing samples end up on both sides and the classifier recognises the recording rather than the fault. A survey across seventeen scientific fields found leakage of this kind in 294 papers, in some cases producing conclusions far more optimistic than the data supported [5].

The shift between operating conditions is itself an active subject. A review of deep-learning work on the CWRU dataset records how widely it serves as a validation reference [6], and a benchmark of four architectures over nine datasets reports generalisation as an open problem of the field [7]. The usual response is a domain-adaptation model that aligns the distributions of a labelled source condition and an unlabelled target condition, for instance by minimising a multi-kernel maximum mean discrepancy across several layers [8]. That line fixes a representation and modifies the model, leaving open what the representation itself contributes to the transfer, which is the question addressed here.

Fault diagnosis in rotating machinery has been addressed repeatedly in this journal, from measurement rather than from model design. Time-domain descriptors such as RMS, energy and entropy have been fed to a feed-forward network to diagnose mechanical faults in spark-ignition engines [9], while temperature and injection pressure have served as measured indicators instead of a signal representation, with a multifactorial ANOVA establishing which factors matter [10]. Earlier, two measurement methods for stator lamination faults were compared on rotating machines without a learned model [11]. A four-level Daubechies db4 decomposition has been applied in the journal to estimate fault distance [12]; the wavelet representation below uses the same family with five levels.

This study measures what happens to four representations when the test load differs from the training load, under a partition designed so that the numbers are not inflated by overlap. The contributions of this work are: (1) a record-level evaluation protocol for the CWRU dataset in which the split is performed by source record and, within a record, by purged contiguous blocks, verified by a leakage check with a negative control; (2) a 4×4 matrix of macro-F1 by (training load, test load) pair for four representations and three classifiers, reported per fold; (3) evidence that no representation holds a consistent advantage across classifiers, the representation by classifier interaction being the largest term of the factorial model; and (4) an ablation, specified in the protocol before the data were split, showing that removing amplitude information does not improve cross-load transfer.

2. Materials and Methods

2.1. Dataset and label verification

Forty drive-end accelerometer records of the CWRU Bearing Data Center [2], sampled at 12 kHz, cover ten class-diameter combinations under four loads (0, 1, 2 and 3 HP, nominally 1797, 1772, 1750 and 1730 rpm). The classes are healthy bearing (Normal), inner race fault (IR), ball fault (B) and outer race fault (OR), the last three at diameters of 0.007, 0.014 and 0.021 inches.

The file naming convention is a known source of error. One file, `99.mat` (Normal, 2 HP), holds both `X098_DE_time`, a duplicate of the 1 HP file, and `X099_DE_time`. Taking the first key ending in `_DE_time` returns the duplicate and labels the 1 HP signal as 2 HP, displacing a column of the results matrix with no visible symptom. The channel is therefore matched exactly against the file number (`X{NNN}_DE_time`), and the script fails rather than falling back to position.

Labels were verified against the physics of the bearing rather than third-party tables. For the SKF 6205-2RS JEM drive-end bearing the characteristic order factors are BPFO 3.5848, BPFI 5.4152, BSF 2.3568 and FTF 0.3983 times the shaft frequency; the rolling-element value of 4.7135 tabulated by CWRU corresponds to 2×BSF, not to BSF. Envelope analysis (band-pass filtering in the 2–5.5 kHz resonance, Hilbert transform, spectrum) confirmed the inner race labels in 12 of 12 records and the outer race labels in 9 of 12, the expected order dominating by a factor of three and five. In the healthy records no characteristic

frequency stood out, an order of magnitude below the fault records. Ball faults required the sideband criterion at 2×BSF modulated by FTF, a defective ball striking both races and crossing the load zone at the cage rate; under it 11 of 12 records exceeded twice the largest peak of the healthy records, the weakest signatures being those at 0.021 in. All 40 records were kept, since discarding those with a weak signature would bias the result upwards, and fault diameter is a covariate for subgroup reporting.

2.2. Windowing and the partition rule

Each record was segmented into windows of 2048 samples with 50 % overlap, yielding 5887 windows. The split is performed by source record, never by window, because overlapping windows of a record share samples.

Outside the diagonal this holds by construction, since training on load i and testing on load j uses different files; on the diagonal it cannot, because the dataset provides a single healthy record per load. One rule was therefore adopted for all sixteen cells: the windows of each record are cut into five contiguous blocks in time and the first window of each internal block is discarded. With a stride of 1024 samples and a window of 2048, windows m and $m + 2$ share no samples, so one discarded window per boundary suffices, removing 160 windows (2.7 %) and leaving 5727. For cell (i, j) and fold k, training uses the blocks other than k from every record of load i, and testing block k from every record of load j.

Off the diagonal the rule leaves record-level separation intact and the blocks only produce five disjoint test sets. On the diagonal, overlap is removed but training and test windows still come from the same record, so the diagonal is a reference and is excluded from every test. The training set is four fifths of one load in all sixteen cells, so no sample-size confound separates the two.

All 80 cell-fold combinations were checked for training windows sharing samples with a test window of the same record, and none was found. The same check without the purge reports 320 violations, which is the 160 overlapping pairs created by four block boundaries in each of the forty records, each counted in the two folds it affects. Table 1 gives the usable window counts; the healthy records last 20 to 40 s against about 10 s for the fault records.

Table 1. Usable windows of 2048 samples with 50 % overlap, by class and load, after discarding the boundary windows between blocks.

Class	0 HP	1 HP	2 HP	3 HP
Normal	233	467	468	469
IR	340	340	340	342
B	341	340	341	341
OR	341	342	340	342

2.3. Signal representations

Four representations were computed from the same windows. Figure 1 shows one raw window per class and Figure 2 shows the same inner-race window under the four representations.

Time-domain statistics. The twelve classical vibration descriptors: mean, RMS, standard deviation, skewness, kurtosis, peak, peak-to-peak, crest factor, shape factor, impulse factor, clearance factor and mean square root amplitude.

FFT magnitude. The real FFT magnitude up to Nyquist, 1025 bins of 5.86 Hz.

Wavelet db4. Five-level discrete decomposition with the Daubechies db4 wavelet, the family used in [12], as the relative energy of each band (six values: the approximation A5 and the details D5 to D1).

STFT spectrogram. The spectrogram in dB, normalised per image to $[0, 1]$, with the transform window set explicitly to $nperseg = 64$: 187.5 Hz per band, 63 frames and a hop of 2.67 ms. BPFO and BPFI are not spectral peaks: the fault excites the 2–5.5 kHz structural resonance and the characteristic frequency appears as the repetition rate of the impacts, along the time axis. At 1797 rpm the impacts occur every 9.3 ms (BPFO) and 6.2 ms (BPFI), so a hop of 2.67 ms places 2.3 to 3.5 frames within each impact period, while a 512-sample window would give 23.4 Hz per band but a hop of 21.3 ms, erasing the impact train. For the same reason the spectrogram enters the comparison as a 16×64 grid of 1024 values rather than a square 32×32 one, which would leave 5.33 ms per column, fewer than two columns per BPFI impact period; the 16×64 grid keeps the 2.67 ms hop and halves the frequency axis to 375 Hz. A 64×64 image of the same transform feeds the convolutional network.

Table 2 summarises the four. The scaler is fitted inside the pipeline, on the training fold only.

Table 2. Dimension, effective resolution and dependence on absolute amplitude of each representation.

Representation	Dim.	Effective resolution	Amp.-inv.	Enters comparison
Time-domain statistics	12	—	no	yes
FFT magnitude	1025	5.86 Hz	no	yes
Wavelet db4 (rel. energy)	6	dyadic band	yes	yes
STFT spectrogram (16×64)	1024	375.0 Hz / 2.67 ms	yes	yes
STFT spectrogram (64×64)	4096	187.5 Hz / 2.67 ms	yes	no

Two of the four discard amplitude by construction, the wavelet returning relative band energy and the spectrogram being normalised per image, while the other two keep the absolute scale that varies with load. Comparing them as they stand would settle any hypothesis about amplitude through the preprocessing rather than the representation, so an ablation was fixed before results were examined: time-domain statistics and FFT magnitude recomputed on windows divided by their RMS value, with SVM-RBF and Random Forest over the twelve off-diagonal pairs. After normalisation the RMS descriptor equals one in all 5887 windows and kurtosis is unchanged.

2.4. Classifiers

Three classifiers that accept vectors were used with all four representations: SVM with RBF kernel, Random Forest with 300 trees and a two-layer perceptron of 128 and 64 units trained for at most 500 iterations, all from scikit-learn [13]. Class weighting was balanced for the SVM and the forest; the perceptron accepts neither class nor sample weights. A standard scaler was fitted on the training fold for the SVM and the perceptron, and omitted for the forest. Hyperparameters were left at the library defaults, so that all four representations meet each classifier under identical settings. A nested search on an inner split of the training blocks was possible without touching the test block; not running one is a limitation, not a constraint of the design.

A two-dimensional convolutional network provides a reference row: three blocks of 32, 64 and 128 channels with batch normalisation and max pooling, then global average pooling and a linear layer, in PyTorch [14], trained on 64×64 spectrograms for a fixed 30 epochs with Adam at 10^{-3} and a batch of 64. The epoch count is fixed rather than set by early stopping, since a validation set would either shrink the training set relative to the other classifiers or select on the test block. The network appears in Table 3 but takes no part in the statistical comparison: it has no counterpart for the other three representations, and an unbalanced design invalidates both Friedman and the ANOVA.

2.5. Experimental design and statistical analysis

The design covers 4 representations $\times$ 3 vector classifiers $\times$ 16 cells $\times$ 5 folds, that is 960 fits, plus 80 for the convolutional reference and 240 for the ablation. The seed was fixed at 42 and results stored per fold. The network ran on an RTX 4050 laptop GPU and the rest on CPU.

The statistical procedure was fixed before the comparison was run, following [15]. The primary test is a Friedman test over the four representations with the twelve off-diagonal pairs as blocks ($N = 12$, $k = 4$), using macro-F1 averaged over the three classifiers and the five folds, with a Nemenyi post-hoc test and a critical difference diagram. The three classifiers share the partition and the signal, so treating each (pair, classifier) combination as an independent block would raise N to 48 and halve the critical distance to about 0.68 on the strength of that dependence. As a robustness check a separate Friedman test was run per classifier, again with $N = 12$, at a Bonferroni-corrected $\alpha = 0.0167$.

A two-way ANOVA on macro-F1 examines representation and load distance (1, 2 or 3 HP of separation; distance 0 is the diagonal and is excluded) with their interaction, over 240 observations, reporting partial eta squared as the effect size. Two exploratory departures accompany it. Those 240 observations are 5 folds within each of 48 cells, and folds of one cell share four fifths of their training windows, so the model was refitted on the 48 cell means. It also averages over the classifier and so cannot test any term involving it, which a three-way model on the 144 (representation, classifier, pair) cell means supplies. The ablation is reported as a paired difference with a bootstrap confidence interval and Cliff's delta, since only two conditions are compared.

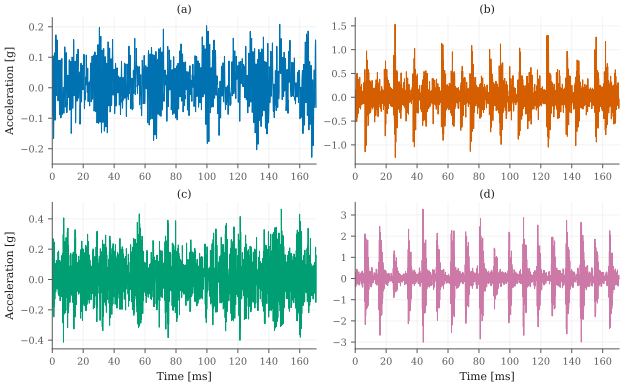


Figure 1. Raw drive-end acceleration signals at 0 HP, one 2048-sample window per class. (a) Normal, (b) inner race, (c) ball, (d) outer race. Each panel has its own y-axis scale: record RMS ranges from 0.066 g (Normal) to 0.425 g (outer race), so a shared scale would flatten three panels.

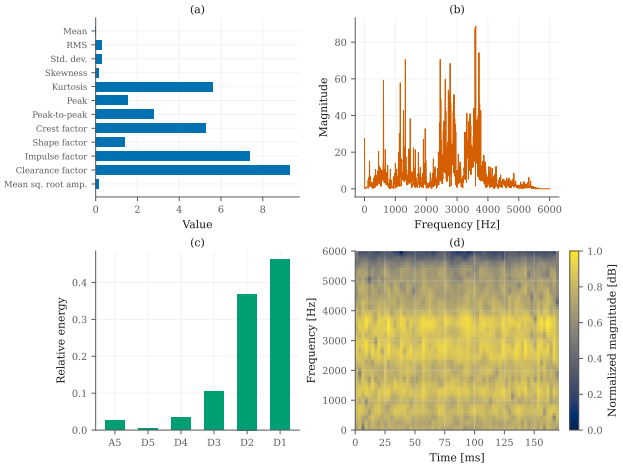


Figure 2. Same inner-race fault window at 0 HP under the four representations. (a) Time-domain statistics, (b) FFT magnitude, (c) db4 wavelet relative energy per band, (d) STFT spectrogram on the 64×64 CNN grid, not the 16×64 grid that enters the comparison. The twelve descriptors of (a) do not share units; wavelet bands run from coarsest approximation (A5) to finest detail (D1).

3. Results and Discussion

3.1. In-domain performance saturates

Within a single load all four representations exceed 0.95 macro-F1, averaged over classifiers: 0.9997 for FFT magnitude, 0.9925 for the spectrogram, 0.9882 for time-domain statistics and 0.9602 for the wavelet, with the breakdown by classifier in Table 3. Under four hundredths separate best from worst and ten of the twelve representation-classifier combinations sit at or above 0.98, so a comparison restricted to the training condition separates little.

Table 3. Macro-F1 (mean ± standard deviation over folds and cells) in-domain and out-of-domain, by representation and classifier. The convolutional network is a reference row and takes part in neither Friedman nor the ANOVA. The best value in each column is in bold, among the twelve rows that enter the comparison; the in-domain column has two rows tied at 1.000.

Representation	Classifier	F1 in-domain	F1 out-of-domain
Statistical	MLP	0.988 ± 0.015	0.799 ± 0.229
Statistical	Random Forest	0.992 ± 0.008	0.869 ± 0.103
Statistical	SVM-RBF	0.985 ± 0.020	0.836 ± 0.174
FFT	MLP	**1.000 ± 0.000**	**0.971 ± 0.043**
FFT	Random Forest	**1.000 ± 0.000**	0.956 ± 0.054
FFT	SVM-RBF	0.999 ± 0.003	0.774 ± 0.061
Wavelet db4	MLP	0.969 ± 0.017	0.830 ± 0.087
Wavelet db4	Random Forest	0.982 ± 0.011	0.892 ± 0.045
Wavelet db4	SVM-RBF	0.929 ± 0.018	0.850 ± 0.074
STFT spectrogram	MLP	0.998 ± 0.003	0.916 ± 0.056
STFT spectrogram	Random Forest	0.980 ± 0.011	0.898 ± 0.069
STFT spectrogram	SVM-RBF	0.999 ± 0.001	0.942 ± 0.054
CNN 2D (reference)	CNN2D	0.971 ± 0.116	0.957 ± 0.119

3.2. Cross-load degradation

Figure 3 shows macro-F1 for every (training load, test load) pair, one panel per representation, diagonal boxed. Averaged over the twelve off-diagonal cells, the drop from the diagonal is 0.074 for the spectrogram, 0.0995 for FFT magnitude, 0.1029 for the wavelet and 0.1534 for time-domain statistics. Resampling the twelve pairs gives intervals of 0.043–0.105, 0.080–0.122, 0.073–0.133 and 0.074–0.240, which all overlap; the pairs share loads, so the intervals are approximate (Section 3.6). Time-domain statistics, which retain absolute amplitude, degrade most, but FFT magnitude, which also retains it and carries no time information, degrades slightly less than the amplitude-invariant wavelet: the ordering follows neither amplitude nor time-frequency content.

Degradation grows with load distance at different rates: at one step the four lie between 0.887 and 0.943 and at three steps between 0.655 and 0.880, most of the widening from time-domain statistics.

The matrix is also asymmetric. With time-domain statistics, training at 0 HP and testing at 1 HP gives 0.795 against 0.678 for the reverse; at maximum separation the two directions give 0.771 and 0.539, the worst cell of the matrix. With statistics and the wavelet, testing at 0 HP is harder in every pair, and the healthy class has 233 windows there against about 468 elsewhere; FFT magnitude shows the opposite asymmetry, so the imbalance alone does not explain it.

The spectrogram is also the most expensive of the four to extract, 5.96 s over the 5887 windows against 0.27 s for FFT magnitude.

The convolutional reference scores 0.971 in-domain, below every vector representation except the wavelet (0.960), and 0.957 out of domain, the highest, a drop of 0.0135. A training failure governs both. Trained at 0, 1 and 2 HP it reaches 0.998, 1.000 and 0.992 in-domain, and 0.892 ± 0.230 at 3 HP; the deficit is one of the five folds at 3 HP, whose four cells fall between 0.396 and 0.489, consistent with the network not converging within the fixed 30 epochs. That fold also produces the standard deviations of 0.116 and 0.119. It is reported as it ran, since the epoch budget was fixed in advance. The row is descriptive and its apparent advantage in transfer should not be read as one.

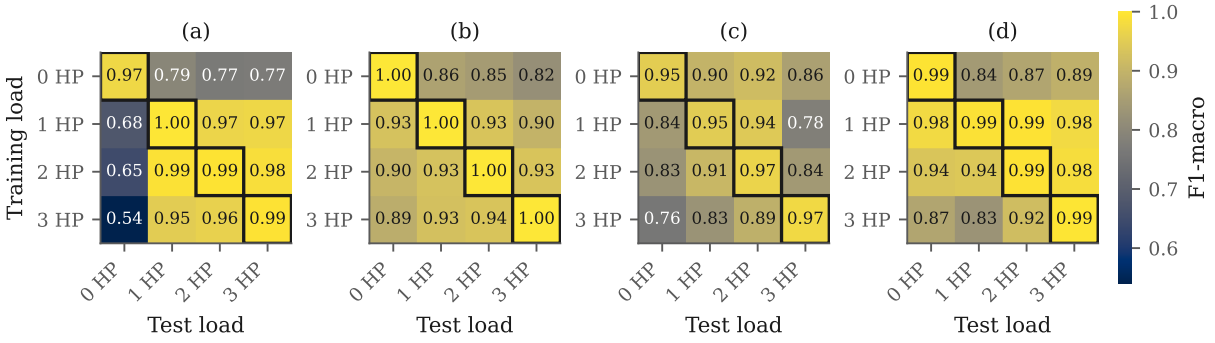


Figure 3. Macro-F1 for every (training load, test load) pair, averaged over the three vector classifiers and 5 folds. Rows are the training load and columns the test load. (a) Statistical, (b) FFT, (c) Wavelet db4, (d) STFT spectrogram. The diagonal is a reference cell, boxed for orientation, and takes part in neither Friedman nor the ANOVA: with a single Normal recording per load it does not admit a record-level split.

3.3. No consistent ranking across classifiers

The primary Friedman test over the twelve off-diagonal pairs does not reject: $\chi^2 = 5.20$, $p = 0.158$, with mean ranks of 1.833 for the spectrogram, 2.500 for FFT magnitude, 2.667 for time-domain statistics and 3.000 for the wavelet, against a Nemenyi critical distance of 1.354. No pair is separated in the critical difference diagram of Figure 4, the smallest post-hoc p-value being 0.119, between the wavelet and the spectrogram.

The three per-classifier tests each reject and disagree on the ordering. The spectrogram ranks first with SVM-RBF ($\chi^2 = 17.10$, $p = 0.00067$), where FFT magnitude ranks last at 3.500; FFT magnitude ranks first with Random Forest ($\chi^2 = 13.70$, $p = 0.0033$, rank 1.333) and with the perceptron ($\chi^2 = 21.00$, $p = 0.00011$, rank 1.083), all three surviving the Bonferroni-corrected $\alpha = 0.0167$. Averaging them for the primary test cancels that disagreement, which is why the omnibus test does not reject while its components do.

The two-way ANOVA of the protocol, over the 240 fold-level observations, returns representation ($F = 15.93$, $\eta_p^2 = 0.173$), load distance ($F = 35.51$, $\eta_p^2 = 0.238$) and their interaction ($F = 5.14$, $\eta_p^2 = 0.119$) as significant at $p < 0.001$ over 228 residual degrees of freedom. Those degrees of freedom assume independent folds, and consecutive folds share four fifths of their training windows. Refitting on the 48 cell means removes the assumption and most of the result: load distance survives ($F = 5.85$, $p = 0.006$), representation falls to $F = 2.62$, $p = 0.065$, and the representation by load-distance interaction to $F = 0.85$, $p = 0.543$. That interaction is an artefact of treating folds as replicates and is not claimed here.

Adding the classifier as a factor locates the effect elsewhere (Table 4). The representation by classifier interaction is the largest term of the model ($F = 4.91$, $p < 0.001$, $\eta_p^2 = 0.214$), above every main effect, while representation by load distance is not significant ($F = 1.76$, $p = 0.113$). The effect of representation depends on the classifier, which the three disagreeing Friedman tests already indicated.

That model treats the twelve pairs as interchangeable replicates within each level of load distance, and they are not: every load appears in six of them. Entering the pair as a block absorbs the dependence, at the cost of load distance, which is a function of the pair and leaves the model. The interaction survives and grows slightly ($F = 5.78$, $p < 0.001$, $\eta_p^2 = 0.223$ over 121 residual degrees of freedom), while the block takes the largest share ($F = 4.87$, $\eta_p^2 = 0.307$): which two loads are crossed matters more than any factor of the design, and load distance does not capture all of it. All models beyond the one specified in the protocol are exploratory and their p-values are uncorrected.

Table 4. Exploratory three-way ANOVA on macro-F1 over the 144 off-diagonal cells, with the five folds of each cell averaged. The diagonal is excluded and the convolutional network takes no part.

Source	SS	df	F	p	η_p^2
Representation (R)	0.1595	3	5.47	0.002	0.132
Classifier (C)	0.0687	2	3.53	0.033	0.061
Load distance (D)	0.2370	2	12.18	< 0.001	0.184
R × C	0.2865	6	4.91	< 0.001	0.214
R × D	0.1030	6	1.76	0.113	0.089
C × D	0.0178	4	0.46	0.766	0.017
R × C × D	0.0336	12	0.29	0.990	0.031
Residual	1.0504	108			

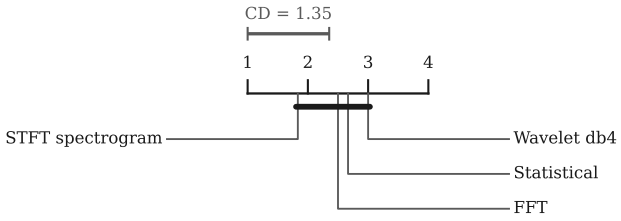


Figure 4. Nemenyi critical-difference diagram for the primary Friedman test ($N = 12$ off-diagonal pairs, $k = 4$ representations, $\chi^2 = 5.20$, $p = 0.158$). Representations joined by a bar are not distinguishable at $\alpha = 0.05$; CD is the Nemenyi critical distance.

3.4. Performance by fault diameter

Over the off-diagonal cells recall is highest for the healthy class (0.994) and lowest for inner race faults at 0.014 in (0.587). The next four subgroups are ball at 0.021 in (0.786), inner race at 0.021 in (0.823), inner race at 0.007 in (0.832) and outer race at 0.014 in (0.849); the remaining four sit above 0.89. Only the first is separated from the rest by a wide margin.

Envelope verification had already identified two of those subgroups, the outer race records at 0.014 in and the ball records at 0.021 in, as having a weak signature. The inner race subgroup at 0.014 in was not, since envelope analysis confirmed all twelve inner race records. The two ask different questions: whether the signature is detectable within a record, and whether a classifier trained on one load recognises it at another. The representations disagree sharply on it, recall being 0.836 with FFT magnitude against 0.485 with the spectrogram, 0.494 with time-domain statistics and 0.533 with the wavelet, so a single subgroup carries part of the representation by classifier interaction.

3.5. Amplitude normalisation does not explain the transfer gap

The ordering in the cross-load results suggests that representations retaining absolute amplitude degrade because amplitude scales with load. The ablation tests it by dividing each window by its RMS value before extracting the two amplitude-dependent representations. No combination improved by an interval that excludes zero, and two degraded. With Random Forest, time-domain statistics lost 0.149 macro-F1 (95 % CI −0.197 to −0.103, Cliff's delta −0.472) and FFT magnitude lost 0.133 (95 % CI −0.151 to −0.116, delta −0.750). With SVM-RBF both intervals cross zero: −0.018 for statistics (95 % CI −0.047 to +0.011, delta −0.205) and +0.033 for FFT (95 % CI −0.002 to +0.067, delta 0.330). That last interval contains zero, so the ablation separates the two classifiers rather than ruling out any benefit: normalisation is harmful with the forest and of uncertain benefit with the SVM.

Removing amplitude therefore does not help transfer with the forest, and the ordering observed above requires an explanation other than amplitude dependence alone.

3.6. Limitations

Each combination of fault type and diameter in the CWRU dataset is a single physical bearing measured under four loads, so crossing loads keeps the bearing fixed and a model may rely on the signature of that specimen rather than on the fault type. The degradation measured here is therefore a lower bound, and external validation on another machine is left as future work.

The diagonal cells cannot use a record-level split, since the dataset provides one healthy record per load. Their purged blocks remove leakage through overlap but not the correlation of sharing a record; they are excluded from every test and the in-domain figures of Table 3 are a reference level, not an estimate of generalisation. That single healthy record also leaves the class imbalanced differently across loads, 233 windows at 0 HP against about 468 elsewhere. Macro-F1 and class weighting mitigate this, but the perceptron accepts no class weights and the direction asymmetry reported above is consistent with the imbalance being part of the cause rather than an effect of load alone.

The envelope spectrum, the reference technique of the field and the one used here to verify the labels, is not among the four representations, which were fixed before the data were touched. It keeps the impact repetition rate and discards the carrier, which makes it the likeliest to transfer across loads, so its absence limits what the comparison covers.

Hyperparameters were left at their defaults, so the per-classifier ranking holds under those defaults only, and the four representations differ in dimension by two orders of magnitude, from 6 to 1025. The two interact: an untuned SVM on 1025 FFT bins is a plausible partial explanation for FFT magnitude ranking last with that classifier and first with the other two. Equalising dimension was rejected because dimension is part of a representation, and a nested search on the training blocks would separate the two readings.

The twelve blocks of the Friedman test are not independent. Each load appears in six pairs, and the pairs (i, j) and (j, i) use the same two sets of records with the roles exchanged, so the exchangeability the test assumes is violated in a direction that cannot be signed without a variance component per load. The blocked ANOVA above measures how much variance the pair carries, but no test here escapes the dependence and, with four loads, no design on this dataset would. The p-value of 0.158 is a failure to separate the four representations, not evidence that they are

equivalent. For the same reason the results are not placed beside published cross-load figures, which are reported under partitions this protocol was designed to avoid: the difference would measure the partition rather than the method.

4. Conclusions

Within a single load the four representations all exceed 0.95 macro-F1 and separate by less than four hundredths, so a comparison under the training condition alone does not identify a preferable one. Once the test load differs the spread widens, from 0.074 for the STFT spectrogram to 0.153 for time-domain statistics, the latter falling to 0.655 macro-F1 at three steps of separation. The bootstrap intervals of those four figures overlap, so the ordering they suggest is not established.

That spread does not translate into a general recommendation. The omnibus Friedman test over the twelve off-diagonal pairs does not reject ($\chi^2 = 5.20$, $p = 0.158$), the three per-classifier tests each reject and place a different representation first, and the largest term of the factorial model is the representation by classifier interaction. FFT magnitude ranks last with SVM-RBF and first with the other two. Representation and classifier should therefore be chosen together, since a recommendation drawn from one classifier does not carry to another.

The amplitude mechanism that motivated the ablation, that representations retaining absolute amplitude degrade because amplitude scales with load, was not supported. RMS normalisation improved cross-load transfer in none of the four tested combinations by an interval excluding zero, and degraded it with Random Forest by 0.149 for time-domain statistics and 0.133 for FFT magnitude. On this evidence, removing absolute amplitude does not buy transfer.

External validation on a different machine would separate the degradation attributable to load from that masked by reusing one physical bearing per fault condition. Subgroup reporting would also repay attention: the inner race subgroup at 0.014 in reached a recall of 0.587 across loads, with the four representations disagreeing by more than three tenths.

Data availability

The CWRU records are public [2]. The analysis code, the per-fold results and the scripts generating every table and figure are deposited in a public repository with a DOI, withheld for double-blind review and provided upon acceptance.

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